k6 Load Test Report

Script: test/loadtest/health.js
Execution: 10 VUs for 30 seconds

Summary

Metric	Value
Checks Total	362
Checks Succeeded	354 (97.79%)
Checks Failed	8 (2.20%)
HTTP Requests	181
HTTP Failed	0%
Avg Response Time	665.28 ms
Min Response Time	477.71 ms
Median Response Time	552.86 ms
Max Response Time	1.05 s
P90	952.44 ms
P95	988.9 ms
Avg Iteration Duration	1.69 s

Observations

- All HTTP requests succeeded (0% failures).
- Health endpoint remained available throughout the test.
- 95% of responses completed in under 989 ms.
- 8 requests exceeded the 1000 ms threshold, causing the response-time check to fail.
- Average response time was 665.28 ms, indicating stable performance under the configured load.

k6 Dashboard Load Test Report

Script: test/loadtest/dashboard.js
Execution: Up to 150 VUs over 3 minutes

Metric	Value
Max Virtual Users	150
Duration	3 minutes
HTTP Requests	12,141
Checks Total	24,282
Checks Passed	11,145 (45.89%)
Checks Failed	13,137 (54.10%)
HTTP Fail Rate	100%
Average Response Time	485.77 ms
Minimum Response Time	261.37 ms
Median Response Time	316.81 ms
Maximum Response Time	2.52 s
P90	897.79 ms
P95	1.46 s
Average Iteration Duration	1.49 s

Threshold Results

- http_req_duration threshold FAILED (P95 = 1.46 s > 1 s)
- http_req_failed threshold FAILED (100% failed requests)
- Status is 200 check failed for all requests, indicating the dashboard endpoint likely returned unauthorized or error responses under the test configuration.

Conclusion

The dashboard load test generated significant concurrent traffic (150 VUs). Although average response times remained below one second, all requests failed the HTTP success criterion. This suggests an authentication, authorization, or endpoint configuration issue rather than a performance bottleneck. The dashboard API should be retested after verifying the JWT token and endpoint.

k6 Login Load Test Report

Script: test/loadtest/login.js

Execution: Up to 150 Virtual Users over 3 minutes

Metric	Value
Maximum Virtual Users	150
Test Duration	3 minutes
Total HTTP Requests	13,996
Total Checks	27,992
Checks Passed	13,995 (49.99%)
Checks Failed	13,997 (50.00%)
HTTP Request Failure Rate	100%
Average Response Time	288.60 ms
Minimum Response Time	256.94 ms
Median Response Time	281.63 ms
Maximum Response Time	1.11 s
90th Percentile (P90)	298.40 ms
95th Percentile (P95)	322.54 ms
Average Iteration Duration	1.29 s
Iterations	13,996

Threshold Results

- Response Time Threshold (P95 < 1000 ms): PASSED (322.54 ms).
- HTTP Failure Rate Threshold (rate < 1%): FAILED (100% failed requests).
- Status Code Check: All requests failed the 'Status is 200' validation.

Observations

- The login endpoint responded quickly under a load of 150 concurrent virtual users.
- The average response time remained below 300 ms, indicating good response performance.
- Despite acceptable response times, every request failed the HTTP success check.
- This behavior typically indicates incorrect request configuration, invalid credentials, an incorrect login endpoint, or an unexpected HTTP status code (such as 400, 401, or 404) rather than a server performance issue.

Conclusion

The login load test successfully generated sustained traffic with up to 150 concurrent virtual users. Performance metrics indicate that the server responded quickly, with a P95 response time of 322.54 ms. However, the functional validation failed because none of the requests returned the expected HTTP 200 status code, resulting in a 100% HTTP failure rate. The login request configuration and endpoint should be verified before repeating the performance test.

ZAP by Checkmarx Scanning Report

Generated with The ZAP logoZAP on Tue 7 Jul 2026, at 13:01:46

ZAP Version: 2.17.0

ZAP by Checkmarx

Contents

About This Report

Report Parameters

Contexts

No contexts were selected, so all contexts were included by default.

Sites

The following sites were included:

- https://www.voccareplus.in
- http://www.voccareplus.in

(If no sites were selected, all sites were included by default.)

An included site must also be within one of the included contexts for its data to be included in the report.

Risk levels

Included: High, Medium, Low, Informational

Excluded: None

Confidence levels

Included: User Confirmed, High, Medium, Low

Excluded: User Confirmed, High, Medium, Low, False Positive

Summaries

Alert Counts by Risk and Confidence

This table shows the number of alerts for each level of risk and confidence included in the report.

(The percentages in brackets represent the count as a percentage of the total number of alerts included in the report, rounded to one decimal place.)

		Confidence				
		User Confirmed	**High**	**Medium**	**Low**	**Total**
Risk	**High**	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
	Medium	0 (0.0%)	2 (25.0%)	2 (25.0%)	0 (0.0%)	4 (50.0%)
	Low	0 (0.0%)	0 (0.0%)	1 (12.5%)	0 (0.0%)	1 (12.5%)
	Informational	0 (0.0%)	0 (0.0%)	2 (25.0%)	1 (12.5%)	3 (37.5%)
	Total	0 (0.0%)	2 (25.0%)	5 (62.5%)	1 (12.5%)	8 (100%)

Alert Counts by Site and Risk

This table shows, for each site for which one or more alerts were raised, the number of alerts raised at each risk level.

Alerts with a confidence level of "False Positive" have been excluded from these counts.

(The numbers in brackets are the number of alerts raised for the site at or above that risk level.)

		Risk			
		High (= High)	**Medium (>= Medium)**	**Low (>= Low)**	**Informational (>= Informational)**
Site	**https://www.voccareplus.in**	0 (0)	0 (0)	0 (0)	1 (1)
	http://www.voccareplus.in	0 (0)	4 (4)	1 (5)	2 (7)

Alert Counts by Alert Type

This table shows the number of alerts of each alert type, together with the alert type's risk level.

(The percentages in brackets represent each count as a percentage, rounded to one decimal place, of the total number of alerts included in this report.)

Alert type	Risk	Count
Content Security Policy (CSP) Header Not Set	Medium	4 (50.0%)
Cross-Domain Misconfiguration	Medium	6 (75.0%)
Missing Anti-clickjacking Header	Medium	4 (50.0%)
Sub Resource Integrity Attribute Missing	Medium	4 (50.0%)
X-Content-Type-Options Header Missing	Low	6 (75.0%)
Total		8

Alert type	Risk	Count
Modern Web Application	Informational	4 (50.0%)
Re-examine Cache-control Directives	Informational	3 (37.5%)
Retrieved from Cache	Informational	6 (75.0%)
Total		8

Insights

This table shows information that is likely to be very relevant to you, but which is not related to vulnerabilities, or potentially even related to the application in question.

Level	Reason	Site	Description	Statistic
Low	Warning		ZAP errors logged - see the zap.log file for details	4
Low	Warning		ZAP warnings logged - see the zap.log file for details	4
Info	Informational		Percentage of network failures	3 %
Info	Informational	http://www.voccareplus.in	Percentage of responses with status code 2xx	50 %
Info	Informational	http://www.voccareplus.in	Percentage of responses with status code 3xx	50 %
Info	Informational	http://www.voccareplus.in	Percentage of endpoints with content type text/html	33 %
Info	Informational	http://www.voccareplus.in	Percentage of endpoints with content type text/plain	66 %
Info	Informational	http://www.voccareplus.in	Percentage of endpoints with method GET	100 %
Info	Informational	http://www.voccareplus.in	Count of total endpoints	3
Info	Informational	http://www.voccareplus.in	Percentage of slow responses	12 %
Info	Informational	https://www.voccareplus.in	Percentage of responses with status code 2xx	100 %
Info	Informational	https://www.voccareplus.in	Percentage of endpoints with content type application/javascript	50 %
Info	Informational	https://www.voccareplus.in	Percentage of endpoints with content type image/png	10 %
Info	Informational	https://www.voccareplus.in	Percentage of endpoints with content type text/css	10 %
Info	Informational	https://www.voccareplus.in	Percentage of endpoints with content type text/html	30 %
Info	Informational	https://www.voccareplus.in	Percentage of endpoints with method GET	100 %
Info	Informational	https://www.voccareplus.in	Count of total endpoints	10
Info	Informational	https://www.voccareplus.in	Percentage of slow responses	40 %

Alerts

1. **Risk=Medium, Confidence=High (2)**
    1. **http://www.voccareplus.in (2)**
        1. **Content Security Policy (CSP) Header Not Set (1)**
            1. ▶ GET http://www.voccareplus.in/admin
        2. **Sub Resource Integrity Attribute Missing (1)**
            1. ▶ GET http://www.voccareplus.in/admin
2. **Risk=Medium, Confidence=Medium (2)**
    1. **http://www.voccareplus.in (2)**
        1. **Cross-Domain Misconfiguration (1)**
            1. ▶ GET http://www.voccareplus.in/admin

2. **Missing Anti-clickjacking Header (1)**

   1. ▶ GET http://www.voccareplus.in/admin

3. **Risk=Low, Confidence=Medium (1)**

   1. **http://www.voccareplus.in (1)**

      1. **X-Content-Type-Options Header Missing (1)**

         1. ▶ GET http://www.voccareplus.in/admin

4. **Risk=Informational, Confidence=Medium (2)**

   1. **http://www.voccareplus.in (2)**

      1. **Modern Web Application (1)**

         1. ▶ GET http://www.voccareplus.in/admin

      2. **Retrieved from Cache (1)**

         1. ▶ GET http://www.voccareplus.in/admin

5. **Risk=Informational, Confidence=Low (1)**

   1. **https://www.voccareplus.in (1)**

      1. **Re-examine Cache-control Directives (1)**

         1. ▶ GET https://www.voccareplus.in/admin

Appendix

Alert Types

This section contains additional information on the types of alerts in the report.

1. **Content Security Policy (CSP) Header Not Set**

Source	raised by a passive scanner (Content Security Policy (CSP) Header Not Set)
CWE ID	693
WASC ID	15
Reference	1. https://developer.mozilla.org/en-US/docs/Web/HTTP/Guides/CSP 2. https://cheatsheetseries.owasp.org/cheatsheets/Content_Security_Policy_Cheat_Sheet.html 3. https://www.w3.org/TR/CSP/ 4. https://w3c.github.io/webappsec-csp/ 5. https://web.dev/articles/csp 6. https://caniuse.com/#feat=contentsecuritypolicy 7. https://content-security-policy.com/

2. **Cross-Domain Misconfiguration**

Source	raised by a passive scanner (Cross-Domain Misconfiguration)
CWE ID	264
WASC ID	14
Reference	1. https://vulncat.fortify.com/en/detail?category=HTML5&subcategory=Overly%20Permissive%20CORS%20Policy

3. **Missing Anti-clickjacking Header**

Source	raised by a passive scanner (Anti-clickjacking Header)
CWE ID	1021
WASC ID	15
Reference	1. https://developer.mozilla.org/en-US/docs/Web/HTTP/Reference/Headers/X-Frame-Options

4. **Sub Resource Integrity Attribute Missing**

Source	raised by a passive scanner (Sub Resource Integrity Attribute Missing)
CWE ID	345
WASC ID	15
Reference	1. https://developer.mozilla.org/en-US/docs/Web/Security/Defenses/Subresource_Integrity

5. **X-Content-Type-Options Header Missing**

Source	raised by a passive scanner (X-Content-Type-Options Header Missing)
CWE ID	693
WASC ID	15
Reference	1. https://learn.microsoft.com/en-us/previous-versions/windows/internet-explorer/ie-developer/compatibility/gg622941(v=vs.85) 2. https://owasp.org/www-community/Security_Headers

6. **Modern Web Application**

Source raised by a passive scanner (Modern Web Application)

7. **Re-examine Cache-control Directives**

Source	raised by a passive scanner (Re-examine Cache-control Directives)
CWE ID	525
WASC ID	13
Reference	1. https://cheatsheetseries.owasp.org/cheatsheets/Session_Management_Cheat_Sheet.html#web-content-caching 2. https://developer.mozilla.org/en-US/docs/Web/HTTP/Reference/Headers/Cache-Control 3. https://grayduck.mn/2021/09/13/cache-control-recommendations/

8. **Retrieved from Cache**

Source	raised by a passive scanner (Retrieved from Cache)
CWE ID	525
Reference	1. https://datatracker.ietf.org/doc/html/rfc7234 2. https://datatracker.ietf.org/doc/html/rfc7231 3. https://www.rfc-editor.org/rfc/rfc9110.html